

Aluna: Pâmela Layla Freitas Barbosa
Professor: Ricardo Taveira
Atividade: 05

1) Explique os conceitos inerentes à implementação de um proxy reverso implementado com o NGINX.

Um proxy reverso é um servidor intermediário que fica entre os clientes (navegadores) e os servidores de origem (aplicações).

Imagine uma central de atendimento de uma empresa. Você liga para o número único (proxy), e eles transferem sua chamada para o departamento certo (backend). Você nunca fala diretamente com o departamento.

Proxy Forward (Proxy Tradicional)	Proxy Reverso
Age em nome do cliente	Age em nome do servidor
Cliente se esconde do servidor	Servidor se esconde do cliente
Exemplo: VPN, acessar conteúdo bloqueado	Exemplo: Balanceador de carga, CDN
Configurado no navegador do usuário	Configurado no servidor

Benefícios do Proxy Reverso:

- Balanceamento de carga: Distribui requisições entre vários servidores
- Segurança: Oculta o IP e características dos servidores backend
- Cache: Armazena respostas para acelerar requisições
- Terminação SSL: Gerencia certificados HTTPS centralizadamente
- Compressão: Compacta respostas antes de enviar ao cliente

Como o NGINX implementa Proxy Reverso:

```
# Define grupo de servidores backend
upstream backend_servers {
    server app1:80;
    server app2:80;
}
```

```
# Configura o proxy
location / {
    proxy_pass http://backend_servers; # Encaminha requisições
    proxy_set_header Host $host;      # Preserva host original
    proxy_set_header X-Real-IP $remote_addr; # IP do cliente
}
```

Prática usando KillerCoda:

- Verificação do ambiente:

```
Editor  Aba 1  Aba 2  +  60 m  [icon]  [icon]
```

```
root@ubuntu:~$ # Verificar nós do cluster
kubectl get nodes

# Verificar pods do ingress controller
kubectl get pods -n ingress-nginx
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
root@ubuntu:~$
```

- Criar aplicações backend

```
Editor  Aba 1  Aba 2  +  60 m  [icon]  [icon]
```

```
name: app1-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: app1
  template:
    metadata:
      labels:
        app: app1
    spec:
      containers:
        - name: nginx
          image: nginx:alpine
          ports:
            - containerPort: 80
          volumeMounts:
            - name: html
              mountPath: /usr/share/nginx/html
      volumes:
        - name: html
          configMap:
            name: app1-html
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: app1-html
data:
  index.html: |
    <!DOCTYPE html>
    <html>
    <body style="background:#e8f4f8; text-align:center; padding:50px">
EOF </html>Pod: <span id="pod"></span></p>lo App 1</p>
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
```

```
root@ubuntu:~$ # Criar service do app1
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Service
metadata:
  name: app1-service
spec:
  selector:
    app: app1
  ports:
    - port: 80
      targetPort: 80
EOF
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
root@ubuntu:~$
```

```

      mountPath: /usr/share/nginx/html
    volumes:
      - name: html
        configMap:
          name: app2-html
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: app2-html
data:
  index.html: |
    <!DOCTYPE html>
    <html>
    <body style="background:#fef9e6; text-align:center; padding:50px">
EOF </html>Pod: <span id="pod"></span><p>lo App 2</p>
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>

```

```

Try: apt install <deb name>
root@ubuntu:~$ # Criar service do app2
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Service
metadata:
  name: app2-service
spec:
  selector:
    app: app2
  ports:
    - port: 80
      targetPort: 80
EOF
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
root@ubuntu:~$

```

- Configurar o Ingress (Proxy Reverso)

```

- path: /app2
  pathType: Prefix
  backend:
    service:
      name: app2-service
      port:
        number: 80
EOF
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>

```

```

Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
root@ubuntu:~$ # Ver ingress criado
kubectl get ingress

# Descrever o ingress
kubectl describe ingress demo-ingress
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
root@ubuntu:~$

```

- Teste 1: Acessar via Ingress

```

root@ubuntu:~$ # Obter IP do ingress
INGRESS_IP=$(kubectl get ingress demo-ingress -o jsonpath='{.status.loadBalancer.ingress[0].ip}')

# Testar acesso ao app1
curl -s http://$INGRESS_IP/app1 | grep -o "APP 1"

# Testar acesso ao app2
curl -s http://$INGRESS_IP/app2 | grep -o "APP 2"
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
root@ubuntu:~$

```

- Verificar rewrite funcionando

```

Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
root@ubuntu:~$ # Acessar com path completo - deve redirecionar para raiz
curl -v http://$INGRESS_IP/app1/teste 2>&1 | grep -E "(GET|Location|HTTP)"
root@ubuntu:~$

```

- Ver configuração do NGINX antes e depois

```
root@ubuntu:~$ # Acessar com path completo - deve redirecionar para
iz
curl -v http://$INGRESS_IP/app1/teste 2>&1 | grep -E "(GET|Location)"
root@ubuntu:~$ # Salvar config atual
kubectl exec -n ingress-nginx $controller_pod -- cat nginx.conf >
ig-com-rewrite.txt

# Editar ingress removendo annotation rewrite-target
kubectl annotate ingress demo-ingress nginx.ingress.kubernetes.io
ite-target-

# Aguardar atualização (30 segundos)
sleep 30

# Salvar nova config
kubectl exec -n ingress-nginx $controller_pod -- cat nginx.conf >
ig-sem-rewrite.txt

# Comparar as configurações
diff config-com-rewrite.txt config-sem-rewrite.txt | head -50
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
```

- Testar balanceamento de carga

```
Try: apt install <deb name>
root@ubuntu:~$ # Ver os pods rodando
kubectl get pods -l app=app1

# Múltiplas requisições podem ir para pods diferentes
for i in {1..6}; do
  echo -n "Req $i: "
  curl -s http://$INGRESS_IP/app1 | grep -o "APP 1"
done
Command 'kubectl' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-1ubuntu0.4)
Try: apt install <deb name>
```